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Organic compounds synthesised in the leaves of a plant can be transported to the plant's roots.

This transport is called translocation and occurs in the phloem tissue of the plant.

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One theory of translocation states that organic substances are pushed from a high pressure in the leaves to a lower pressure in the roots.

Describe how a high pressure is produced in the leaves.

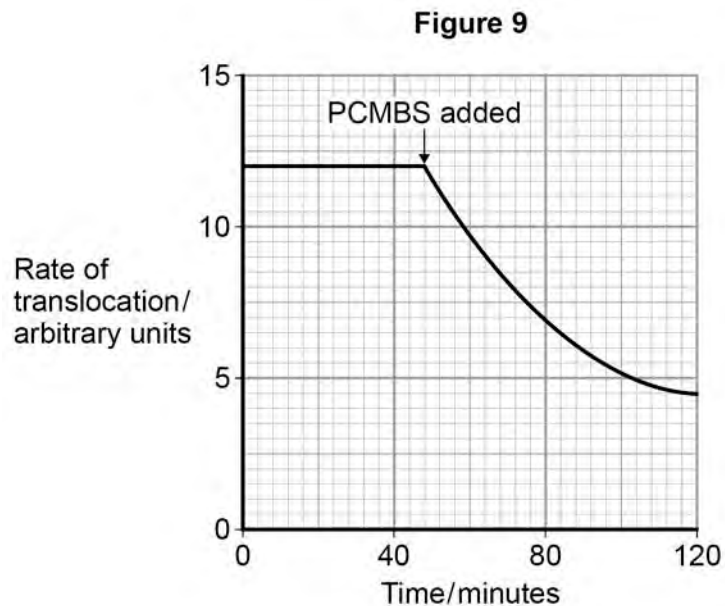
[3 marks]

[Extra space]

Question 9 continues on the next page

PCMBS is a substance that inhibits the uptake of sucrose by plant cells.

Scientists investigated the effect of PCMBS on the rate of translocation in sugar beet. **Figure 9** shows their results.



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During their experiment, the scientists ensured that the rate of photosynthesis of their plants remained constant. Explain why this was important.

[2 marks]

- 0 9 . 3** The scientists concluded that some translocation must occur in the spaces in the cell walls.
Explain how the information in **Figure 9** supports this conclusion.

[2 marks]

Turn over for the next question